

Intelligent reed solomon codec algorithm for digital radio and television media broadcasting

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ABSTRACT

Data dependability and error correction are ensured by the widespread usage of Reed-Solomon codes (RSC) in massive data storage and transmission systems. They shine while dealing with burst faults, which are prevalent in many transmission and storage settings and cause corruption of many consecutive bits. Developing technologies for voice communication in the shortwave frequency is an important topic of research. A top priority for these kinds of communication networks is to make them more resistant to noise, which will extend their range, decrease their power consumption, and ultimately make their user devices smaller and lighter. A method that uses voice codec to transform speech into bits and back again is the most effective way to ensure that voice communication lines in the current world are as noise-free as possible. Here, lowering the likelihood of a bit mistake at the output of the error correcting code is the primary means of improving the noise immunity of these communication systems. In order to minimize the likelihood of a bit error for a given signal-to-noise ratio along with channel model variables for Additive White Gaussian Noise (AWGN) and multipath Rayleigh channels, this work aims at creating a methodology for determining the settings of error correcting codes, synchronization sequences, while direct spread sequence spectrum signal codes. So, techniques like Forward Error Correction codes need to be put in place to deal with these problems properly. In order to rectify broadcasts, we propose in this work to use wireless codes on a variety of deployed radio and Television (TV) equipment using Convolutional Neural Networks (CNNs). We assess this approach in comparison to the Reed-Solomon codes and examine the novel RS codes. We show in our simulations that running the RS codes via the MATLAB simulator yields better results.

1. Introduction

In a digital broadcasting system, which includes a digital broadcasting station, a group of digital broadcast receivers, while a switched network, the system works as follows: the digital signal is sent from the digital broadcasting station to the group of digital broadcast receivers, while the receivers share error correction information through the network in order to make up for any local reception errors at each location. Digital broadcasting is the overarching subject of this innovation [1,2]. To be more specific, it's about digital broadcast receivers that are part of digital broadcasting systems. In these systems, broadcast stations use digital signals for modulating an analog carrier signal before sending it out over radio or cable. At the other end of the transmission chain, digital broadcast receivers digitally demodulate the received signal, identify and fix mistakes, and finally decode the result back into the initial digital signal [3,4]. DVB-T (Digital Video Broadcasting - Terrestrial) ensures robust reception of digital TV signals by using

interleaving convolutional codes and Reed-Solomon codes [5]. DAB (Digital Audio Broadcasting) employs error-correcting methods to provide high-quality digital audio, regardless of the network connection or device used [6]. HD (High Digital) Radio is a digital radio standard that offers data and audio services with little distortion by use of error correction [7]. DVB-SH-A ((Digital Video Broadcasting – Satellite services to Handhelds) is mobile TV standard that uses turbo coding and other error correcting methods [8].

Various data compression formats are often used to lower the transmission bit-rate of digital signals so that they may fit within bandwidth constraints or allocations [9]. Channels are given to each coded digital transmission, which modulates the corresponding band of an analog carrier signal for broadcasting over cable or radio. But local physical phenomena like attenuation, cross-talk, along with dynamic multipath interference may severely distort signals in radio broadcasts across both land and space. The same issues with signal interference [10], particularly cross-talk caused by unshielded conductors' magnetic

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fields, may also arise in cable transmissions. Digital broadcast receivers get bits wrongly demodulated from the received analog signal because to these aberrations [11,12]. The decoding and decompression processes of a digital signal greatly amplify the effects of single bit mistakes if they are left unchecked. The data can only go in one way in a digital broadcast, from the broadcast station to the receivers, thus if there are any mistakes, the receivers can't ask for the original data to be retransmitted [13–15].

To overcome this issue, digital broadcasting standards outline forward error correction systems, which basically provide data redundancy by converting it into parity data. This makes the transmission more error-resistant. Receivers are able to identify and fix mistakes, as well as restore the original digital signal, thanks to the parity data. Forward error correction doesn't always work, and receivers that have more mistakes than parity data can't fix any of them. The standards also outline digital streaming formats for arranging and transmitting the signal data in blocks, with each block containing a separate portion of parity data. This way, in the event of a burst of errors, data can only be erased from a small number of blocks, while the rest of the received digital signal is preserved. This helps to divide failures and maintain quality-of-service. On the other hand, digital signal reception could be entirely disrupted if a significant sustained bit-error rate prevents decoding of several blocks. Here, we receive and demodulate certain bits successfully, but we don't have enough data to decode the (compressed) digital signal, so we just throw away the bits we got. So, the average bit-error rate experienced by local receivers essentially limits the coverage region of a digital broadcast. Current methods for extending this coverage include boosting the strength of the broadcast transmission to increase the signal-to-noise ratio at nearby receivers, increasing the amount of parity information or coding redundancy to make the system more error-tolerant, or constructing transmission towers with repeaters to cover the periphery. The implementation of these techniques is quite expensive and does not scale well, unfortunately. Due to the hemispherical dispersion of the broadcasting power, the received signal strength is inversely linked to the square of the distance from the transmitter; hence, to keep the signal quality same while tripling the distance, the initial transmission power must be quadrupled. While increasing the amount of parity data in a broadcast makes reception possible at lower SNRs (Signal Noise Ratio) and increases the broadcast range, it also decreases the true bandwidth of the transmission route. Lastly, there is a substantial investment in time, effort, and money needed to construct repeater transmission towers, and their development might cause even more interference with signals. We thus need a less expensive means of enhancing the coverage area and quality of service of digital broadcast transmissions. By outlining a digital broadcast receiver that, when transmitted with other digital broadcast receivers, can correct for independent bit errors, the current invention circumvents the shortcomings of the previous art.

The purpose of this innovation is to address the issue of localized bit errors in digital broadcast reception by developing, transmitting, receiving, and decoding digital messages routed via a switched network. A digital broadcast receiver that can locate other digital broadcast receivers on a switched network, establish and maintain online communication with them, and negotiate data exchange with them is another goal of the current invention. To further improve the chances of other digital broadcast receivers being able to correct errors and to reduce the channel selection latency when switching among the decoded outputs of multiple digital signal channels, the current invention also aims to create a digital broadcast receiver capable of decode and buffer multiple channels from a digital broadcast at the same time. The digital broadcast structure that codes digital data while adds extra data for control, data transfer, while error correction is the first embodiment of the present invention. The digital signal is then transmitted to a number of digital broadcast receivers via radio or cable. The following components make up a digital broadcast receiver in one implementation: a demodulating circuit that converts an analogue signal into a digital signal; a detecting

circuit that detects transmission errors in the received digital broadcasting signal; a protecting circuit that stores the received digital broadcasting signal temporarily; a communicating circuit that sends and receives data packets in a switched network; while a correcting circuit that fixes the received digital broadcasting signal.

Better spectrum efficiency, more resistance to noise and interference, and improved signal quality have resulted from terrestrial TV broadcasting's transition from analog to digital technology throughout the last 20 years. An improvement to the viewing experience has resulted in the introduction of High Definition Television (HDTV). Having a bit error rate as low as 10–12, or fewer than one uncorrectable mistake throughout one hour of continuous transmission of 5 Mbps information stream, is a problem that HDTV while Ultra High-Definition Television provide. To achieve such minimal mistake rates, one must optimize service operation, design the network meticulously, and use strong standards.

In order to guarantee dependable data transmission even when noise and interference are present, digital television and radio systems use error correcting algorithms. By incorporating redundancy into the sent data, these strategies enable the receiver to identify and rectify transmission mistakes. Reed-Solomon codes, Hamming codes, while convolutional codes are all common ways for correcting errors; each of these codes has its advantages and disadvantages. To provide reliable data transmission, Digital Video Broadcasting – Terrestrial (DVB-T) employs techniques such as interleaving convolutional codes and Reed-Solomon codes. Whether it's audio or video, digital broadcasting depends on data transfer reliability. The use of error correction codes is critical in reducing the impact of transmission-related signal corruption caused by noise and interference. In order for the receiver to detect and fix mistakes, Error Correction Codes (ECC)s supplement the initial signal with redundant data. In order to circumvent typical transmission deficiencies, this redundancy is engineered to be adequate. The Key Techniques are Reed-Solomon (RS) codes which can be useful for fixing burst problems, often used in DVB-T. Convolutional codes are used in conjunction with interleaving to address burst problems; used for fixing bit errors in continuous channels. Hamming code is used in memory and basic communication systems, which are good candidates for this kind of error correction, which involves fixing individual bits. Turbo codes used as an internal code in hierarchical modulating systems, they provide excellent error-correction capabilities.

In order to guarantee dependable data transfer and playback in digital radio, television, as well as new media, error correction is essential, even when there is a chance of signal interference or transmission faults. Methods like Forward Error Correction with error-detecting codes help accomplish this. Error Correction Methods are Forward Error Correction (FEC) enable the receiver to identify and correct transmission errors without retransmitting the data, FEC augments the original signal with redundant data. Error-detecting codes do not fix the received data, they do indicate that retransmission or another error handling technique is required. Concatenated Codes provide strong error correction, some systems use a combination of error-correcting codes. One such standard that employs a concatenated FEC technique is Digital Video Broadcast Terrestrial. This system incorporates both convolutional and RS codes. Interleavers are burst errors, which occur in consecutive bits and may be caused by interference, are less likely when using these strategies, which reorder the data bits before transmission.

Compared with the state-of-the-art RS decoder, the proposed decoder improved throughput by 150 % and reduced average latency by 81.6 %, according to the hardware implementation results.

By measuring the number of errors in the received code sequence, this research hoped to enhance the performance of real-time communication, eliminate decoding redundancy, and dynamically alter the decoding parameters.

This section provides a brief overview of the paper's key points.

- (1) We provide an RS decoding method that adapts to the channel over time using channel soft information. In order to decrease decoding redundancy with decoding latency, it assesses the time-varying channel setting and produces an appropriate number of TVs and syndromes.
- (2) We present an adaptive RS decoder with outstanding performance that works with time-varying channels. Both the hardware efficiency and the decoder's decoding latency are substantially improved.

Below is the remaining portion of the document. The traditional RS decoding technique is shown in [Section 2](#). [Section 3](#) introduces the RS algorithm. A revised time-varying channel adaptive RS method and a suggested RS decoder are introduced in [Section 4](#). The outcomes of the execution are detailed in [Section 5](#). The conclusion is reached in [Section 6](#).

2. Related work

By including a return channel via the Internet, the researchers of [16] context data broadcasting may progress towards the worldwide idea of interactive multimedia transmission by satellite. Reporting on RAI (RadioActive Iodine) trial services conducted on the Hot Bird 1 satellite, the paper outlines a potential system architecture and identifies potential possibilities for the deployment of interactive multimedia services by satellite.

In order to improve the transmission systems' error correction performance, researchers working on digital terrestrial television broadcast standards as ATSC 3.0 (Advanced Television Systems Committee 3.0) and DVB-T2 (Digital Video Broadcasting – Second Generation Terrestrial) used LDPC (Low-Density Parity-Check) codes with extended codeword lengths [17]. Extended LDPC codeword lengths might be advantageous for 5 G NR-based broadcast services, which may be included in an upcoming 3GPP (Third Generation Partnership Project) release, much as FeMBMS (Further evolved Multimedia Broadcast Multicast Service). It is important to thoroughly examine the current coding processes before adding additional codeword lengths to the 5 G NR standard. In order to achieve longer LDPC codes, that article first describes the 5 G NR LDPC coding chain and then proposes a series of small modifications to the 5 G NR standard. They compare the possible performance of these enhancements to that of the LDPC codes specified in the ATSC 3.0 specification and assess their usefulness.

In their review of ATSC 3.0, the next-generation digital terrestrial broadcasting standard, the authors of [18] provide a high-level summary of the physical layer specification. Using waveforms based on orthogonal frequency division multiplexing and robust low-density parity check forward error correction codes, ATSC 3.0 is compatible with current and future ATSC standards without imposing any backwards-compatibility constraints. Nonetheless, it brings a slew of innovative technical features, like 2-dimensional non-uniform constellations, enhanced along with ultra-robust LDPC codes, power-based layered division multiplexing for effectively providing fixed and mobile services on the same RF channel, and a new scheme for frequency pre-distortion multiple-input single-output antennas. Additionally, ATSC 3.0 permits the use of dual-polarized multiple-input multiple-output antenna systems, the bonding of two RF channels to boost the service peak data throughput, and the exploitation of inter-RF channel frequency diversity. In addition, ATSC 3.0 offers a very versatile data multiplexing method that uses the power, frequency, and time dimensions, as well as a ton of configuration options (such two time interleavers, twelve guard intervals, sixteen pilot patterns, and six modulation orders, to name a few).

Transmission mistakes caused by frequencies selective fading, channel noise, and other propagation phenomena are the focus of the experimenters in [19]. Digital terrestrial television broadcasting uses the DVB-T2 standard, which stands for second generation terrestrial digital

video broadcasting. It is the DVB consortium that issues that standard. Digital cable TV while cable modem applications often use 256-QAM. A wideband SDR platform is to be used to implement and test the modulator and demodulator in that suggested system in accordance with the standard.

Reducing the likelihood of a bit mistake at the output of the error correcting code is the primary method to enhance the noise immunity of these communication systems, according to the authors of [20]. With a given signal-to-noise ratio while channel the model's settings for AWGN (Additive White Gaussian Noise) and multipath Rayleigh channels, that work aims to construct error correcting codes, synchronization sequences, while direct spread sequence spectrum signal codes with the minimum probability of bit errors possible.

That letter proposes and demonstrates the writers of [21]. Quadrature amplitude modulation (QAM) symbol bit error ratio, forward error correction protection capabilities, and quantization bits all come together in a joint design. The most important quantization bits in Two-Level Coding (TLC) in Digital Radio-over-Fiber systems are allocated to the least trustworthy bits of modulated QAM sequences and safeguarded by a FEC (Forward Error Correction). On the other side, the more trustworthy bits of QAM symbols are given the less important bits that do not have FEC protection. A conventional single-mode fiber transmission of 11 km is used in the experiments to assess the performance. Based on the results, the π -RoF based on TLC achieves nearly identical efficiency under the 0.34 % error vector magnitude thresholds of 65536QAM wireless signals, with a complexity reduction of 54.55 % for 16QAM and 67.66 % for 64QAM optical transmissions, respectively, as compared to the conventional bit-interleaved coded modulation about all bits encoded and a maximum iteration number of 2.

Using data collected after RS decoding while C/N information for every carrier, the experts in [22] suggested a way to enhance the receiver's error correction capabilities during Viterbi decoding. They also detailed the outcomes of their simulations testing the efficacy of that method.

A steganography method that allows for recipient-side error detection and repair is proposed by the researchers in [23]. Separate chunks of 2times are rationally cut from the original picture. In all, four quotients and four least significant bits are produced from these four pixels. The seven most significant bits of a pixel's decimal value make up each quotient. It is possible to conceal 8 bits of data in each block. A modified Hamming algorithm is used to calculate four superfluous bits from the eight bits of data. Quotient value differencing (QVD) or bit substitution hides the eight data bits plus one superfluous bit in the quotients. When quotients use QVD for camouflage, the indication bit is set to 1. If bit substitution is used for quotient camouflage, then the indication bit is set to 0. The LSBs (Least Significant Bits) of the four pixels hold the indication bit, three more superfluous bits, and themselves. It is possible to extract data from a block and use an error correction process on the receiver side to fix a 1-bit error over all 8 bits of data. It is possible to identify and fix faults in the received data at the receiver without decreasing HC or increasing distortion, according to the experimental findings.

The authors of [24] provide a comprehensive look of the history, design, and use of Error-Correcting Codes (ECC) in a variety of contexts. ECC may improve data correctness and system dependability by detecting and fixing mistakes that occur during data transmission as well as storage. In recent years, ECC technology has become more important, especially in fields like data storage, digital communication, and wireless networks. With an emphasis on recent developments, the article mainly covers the following codes: Polar codes, Low-Density Parity-Check (LDPC), Turbo codes, Reed-Solomon codes, and Hamming codes. It provides Software-Defined Error-Correcting Codes (SDECC), that combine software and hardware technologies to offer more versatile and effective error handling solutions, and examines the principles of conventional ECC methods, focusing on their shortcomings when dealing with double-bit errors, using LDPC as an example. In addition,

by combining different traditional ECC methods for experimental validation, it shows that SDECC is much better at fixing system crashes and recovering data errors, especially in big data settings, where they can spread data errors and failures. The results of that study provide fresh insights and possible avenues for ECC technology's future advancement.

When faced with software and hardware faults or incorrect configuration data given by broadcasters, DTV receivers might stop operating. The authors of the study that was presented in [25] address that issue and suggest a mechanism for self-diagnosis and behavior correction.

3. Reed solomon coding

For non-binary coefficients, this research used the decimal class of BCH(Bose–Chaudhuri–Hocquenghem) codes, namely the RS coding. A decimal number, the magnitude of the mistake in RS coding depends on the Galois field. Aiming to use simulated code data in the simulated channel propagation model, it comprises modules for source coding, channel coding, modulation, demodulation, data storage, and an abundance of channel simulation.

The Galois field $GF(2^m)$, where m is the total amount of bits in a symbol, is where RS codes are defined. A set of symbols that represent $GF(2^m)$ are $\{0, 1, 2, \dots, (2^m - 1)\}$. After adding the parity symbols to an information message coded using the RS code that begins with the symbol k , a codeword containing the symbol $n = (2^m - 1) = k + 2t$ is attained, besides denoted through $RS(n, k)$. The $RS(n, k)$ code allows for the correction of up to t erroneous symbols if the parity symbol length is recognized as $2t$. The following is the information polynomial for data with symbols of length k :

$$b(x) = b_{k-1}x^{k-1} + \dots + b_1x + b_0 \quad (1)$$

Eq. (2) shows how the minimum polynomial systematic $RS(n, k)$ code is used to construct all of the b_i coefficients, where $b(x)$ is an element of the Galois field.

$$c(x) = b(x)x^{n-k} + p(x) \quad (2)$$

$p(x)$ standing for the parity polynomial, which is the $(n-k-1)$ degree leftover after dividing the $(n-k)$ times shifted information polynomial by the generator polynomial $g(x)$.

$$p(x) = [b(x)x^{n-k}] \bmod g(x) \\ g(x) = (x + \alpha^1) + (x + \alpha^2) + \dots + (x + \alpha^{2t}) \quad (3)$$

The $(n-k)$ degree $g(x)$ polynomial takes coefficients within $GF(2^m)$ besides is fully divided through x^{n-1} . The $RS(n, k)$ phrase, which is constructed using the formula (5); the symbols represent components of $GF(2^m)$

$$c = (c_{n-1}, c_{n-2}, \dots, c_2, c_1, c_0) \quad (4)$$

The communication channel transmits the c codeword to the recipient. In Eq. (6), the receiver takes the new codeword as $r(x)$ when the channel experiences noise or distortion, such as $e(x)$.

$$r(x) = c(x) + e(x) \quad (5)$$

Error identification and repair from data conveyed across a noisy channel is what RS decoding is all about. Although more involved than encoding, this section will provide a concise overview of the decoding process. For more details, see and. In order to retrieve the rectified codeword of $c(x)$, RS decoding attempts to fix the wrong symbols in the $r(x)$ message. The calculation of the received signal's syndrome (error indicator) is the initial step in locating the message's problem.

$$S(\alpha^1) = r(\alpha^1) = e_{p1}(\alpha^1)^{p1} + e_{p2}(\alpha^1)^{p2} + \dots + e_{pt}(\alpha^1)^{pt} \\ S(\alpha^2) = r(\alpha^2) = e_{p1}(\alpha^2)^{p1} + e_{p2}(\alpha^2)^{p2} + \dots + e_{pt}(\alpha^2)^{pt} \\ S(\alpha^{2t}) = r(\alpha^{2t}) = e_{p1}(\alpha^{2t})^{p1} + e_{p2}(\alpha^{2t})^{p2} + \dots + e_{pt}(\alpha^{2t})^{pt} \quad (6)$$

For this condition, there are two t -terms, such $S(\alpha^1), S(\alpha^2), \dots, S(\alpha^{2t})$.

The $e_{pu}(\alpha^i)$ denotes the error's location, while the syndrome variable denotes the amplitude of the error at the p_u position. Variable i and the number of errors, u , are defined as $1 \leq u \leq t$ and $1 \leq i \leq 2t$, respectively. If the $2t$ pieces of $S(\alpha^i)$ are all zero, then the result is correct. Therefore, a correction is unnecessary. Nevertheless, error location polynomials need to be discovered if any of the $S(\alpha^i)$ terms are nonzero. Polynomial to detect errors in $x = \alpha^i$ is exposed as,

$$\Delta(x) = \Delta_0 + \Delta_1x + \dots + \Delta_{u-1}x^{u-1} \quad (7)$$

The most well-known way to generate an error location polynomial in RS coding is the Berlekamp-Massey technique. The simplest approach to determining the roots after using this procedure to create the polynomial (α^i) of the polynomial defined in $GF(2^m)$ requires using the Chien search technique. Following the discovery of the polynomial's roots, the inverse roots $(x^{-1} = (\alpha^i)^{-1})$ of these roots in $GF(2^m)$ give the position of the error.

Although the error location polynomial, roots, and error sites were discovered in the first step, this information is insufficient for error correction since non-binary integers are suggested to be encoded using RS. Hence, it is also necessary to identify the magnitude of the mistake. Equation () is used to produce the error amplitude polynomial $\Omega(x)$, which is used to calculate the error's amplitude.

$$\Omega(x) = [S(x)\Delta(x)] \bmod (x^{2t}) \quad (8)$$

To determine the magnitude of the mistake, one uses the Forney equations to build the polynomial of the error amplitude. In terms of the value of, the magnitude of the mistake α^i in $GF(2^m)$ is

$$e_{pu} = \frac{\Omega(\alpha^{-i})}{\Delta'(\alpha^{-i})} \quad (9)$$

where $\Delta'(x)$ indicates the derivative of $\Delta(x)$. As a result, the receiver's noisy $r(x)$ codeword contains the precise location and magnitude of the u locations' faults. The information message's error in the $RS(n, k)$ codeword is fixed according to Equation (), where $\oplus$ indicates a bit-wise ex-or operation, after every error value ($e(x)$) have been identified.

$$\hat{c}(x) = r(x) \oplus e(x) \quad (10)$$

4. Proposed work

For error correction and dependable data transmission across many platforms, including television, digital radio, and IP-based services, RS codes are extensively used in digital broadcasting and new media. To improve the quality and durability of the received signals, the RS code is often used with other coding methods, such as convolutional codes, to reduce the impact of transmission noise and interference. The receiver is able to identify and fix transmission faults because these block codes provide redundant parity signals to data. When dealing with fading or noisy channels, as is common in wireless broadcasting, this becomes much more crucial.

There is a total of n symbols in an RS encoded message, with m bits per symbol. Out of these, k symbols include the original uncoded information and the rest are the additional parity. Fig. 1 shows an encoded message sample.

Here is how an RS system defines the highest value of detected and correctable message errors:

$$t = \frac{n - k}{2} \quad (11)$$

Given the present state of big data, the technological methods for educating radio and television directors are out of date and cannot keep up with the demands of contemporary development. Without medium loss, radio wave signals flow in a perfectly symmetrical and isotropical space. In this form of space, radio waves travel unimpeded by medium

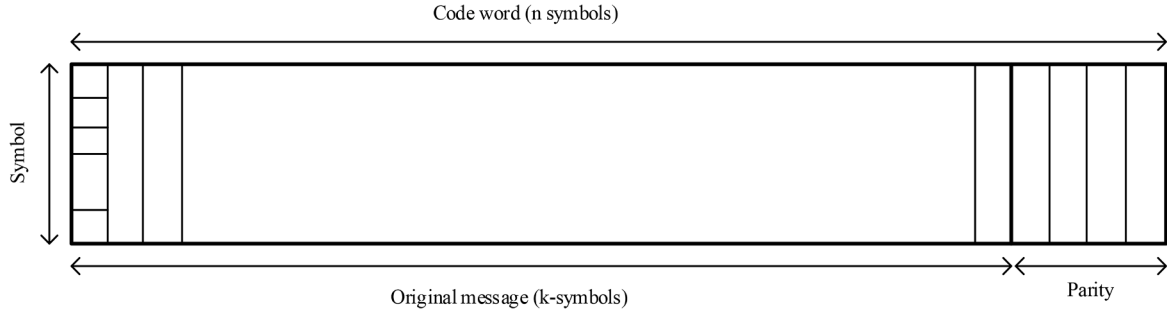


Fig. 1. Traditional RS codec model.

and boundary conditions; the only effect of spatial diffusion is a drop in power flow density with increasing distance. The free space model is applicable at any frequency. Here is a definition of the fundamental transmission loss in open space:

$$L_f = 32.4 + 20\log f + 20\log d \text{ dB}$$

where :

$$f : \text{ frequency (MHz)} \quad (12)$$

$$d : \text{ distance (km)}$$

Field strengths in the free space model may be computed in the following way when omnidirectional transmitting power signals are required:

$$E = P_t - 20\log d + 74.8\text{dB}(\mu \text{ V/m})$$

where :

$$P_t : \text{ omnidirectional transmitting power (dB(W))} \quad (13)$$

$$f : \text{ frequency (GHz)}$$

$$d : \text{ distance of radio wave line (km)}$$

Using a frequency of 700 MHz and a transmitting power of 30 dBW, the correlation between fundamental transmission loss, distance, predicted field strength, and free space distance is shown in Fig. 2.

To create a codeword that can fix a limited number of mistakes, error-correcting codes add redundant bits or symbols to the protected

data. Specialized in symbol-oriented linear block codes, we consider generalized Reed-Solomon codes $RS(q;n,k)$. A finite field with q elements may have k data symbols encoded using these codes $\mathbb{F} := \mathbb{F}_q$ serving as a codeword with n elements. The space is a vector subspace, and the only acceptable n -element sequences are called words $\mathbb{F}^n$ inside each block, which consists of n -element sequences. We say that a code is its collection of codewords, and it's possible for a single code to work with many decoding and encoding methods.

A minimum distance of $d \leq n - k + 1$ characters is required between each codeword in each (n,k) code C . Every decoding method has a threshold when dealing with symmetric channels.

$$t \leq \lfloor (d-1)/2 \rfloor \leq \lfloor (n-k)/2 \rfloor \quad (14)$$

it can properly decode up to a certain maximum amount of symbol mistakes. What this means is that adding $2t$ symbols is the minimum required to reliably fix t symbol issues. Unless stated otherwise, we shall proceed with $t = \lfloor (d-1)/2 \rfloor$.

However, it is possible to fix a certain amount of symbol mistakes by designing a beyond-bound decoder. Even if the likelihood of failure is minimal, such a decoder will always fail to fix all patterns of u symbol mistakes; this is because certain blocks will be within u distance of two codewords in C .

When the channel model favors mistakes impacting the l symbols in

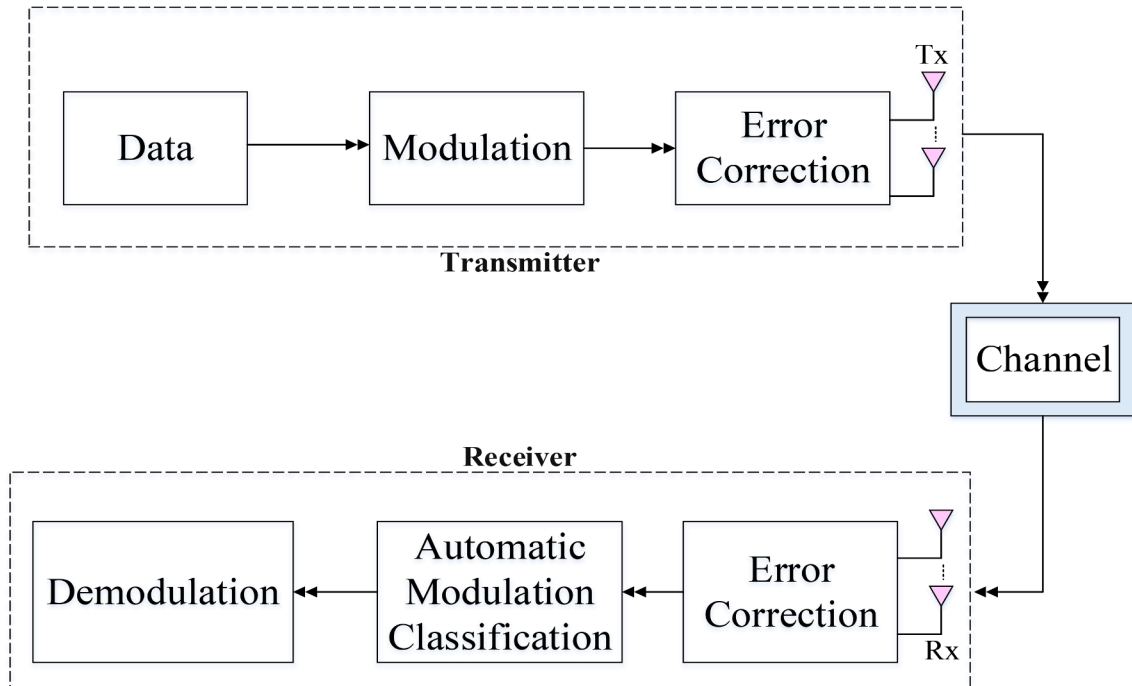


Fig. 2. Proposed error correction model.

one column simultaneously (e.g., failures affecting a channel, a gadget, etc.), IRS codes enable beyond-bound collaborative decoding. By keeping in mind that all of the codes' essential equations must be satisfied by a single degree-e locator, the simplest version of this technique is able to fix e column errors up to $E = \lfloor e \rfloor$ symbol errors. We get $N-K-E$ affine constraints for the e -unknowns from this. When $e \leq \lfloor (N-K)/(l+1) \rfloor$, the system is probable to possess a unique solution. However, after we pass the Komamiya-Singleton constraint, we cannot guarantee uniqueness and decoding becomes probabilistic. Alternatively, a multiple shifting register synthesis approach may be used, which is more efficient, to find solutions through solving these linear equations.

By using more intricate methods like power decoding, decoding may be significantly further extended. Decoding may be done probabilistically up to a radius of

$$R(n, k, \ell) := \left\lfloor n \left(1 - (k/n)^{\ell/(\ell+1)} \right) \right\rfloor \quad (15)$$

elements in a temporal complexity that is quite small. This decoder is intriguing in other settings, but it's too sluggish for memory systems.

Using the received bit-level value $r(i, j)$, whereby i is the symbol position as well as j is the bit position, the error symbol number as well as unreliable value of the codeword are assessed in step 1, as previously said. For each codeword, we tally all the bits where bit-level value is lower than the threshold $Th1$, which is set at 0.3 in this article. The unstable bit position group Δ is updated after selecting the lowest bit γ_i for each symbol [26].

The second step is to set up the adaptive adjustment for the syndrome and the TV. To begin, the codeword is valid if and only if all of the syndromes computed from the initial test vector $TV1$ are zero. On the other hand, mode 1 is executed if the counter, which represents the unreliable bit number, is smaller than $Th2$. In order to differentiate between $e \leq t$ and $e > t$, the total amount of bits whose level value are smaller than $Th1$ is called the threshold $Th2$. Here, e represents the amount of error symbols.

It is possible to use hard decision decoding if the counter is less than or equal to $Th2$, which means that the total amount of error symbols is less than or equal to t . If not, RSC decoding may be used. $Th2$ may be seen as data equivalent to $2t$, supposing that decoding performance is guaranteed. $Th2$ values may be 16 for the usual (255,239) RS codes, as indicated in Table 1. Increasing the value of $Th2$ is suitable if some efficiency is sacrificed for faster decoding. The following equations are satisfied by the syndrome S , the error location H , and the error value δ [27].

$$\begin{aligned} S_0 &= \delta_1 \beta_1 + \delta_2 \beta_2 + \delta_3 \beta_3 + \dots + \delta_e \beta_e \\ S_1 &= \delta_1 \beta_1^2 + \delta_2 \beta_2^2 + \delta_3 \beta_3^2 + \dots + \delta_e \beta_e^2 \\ &\vdots \\ S_{2t-1} &= \delta_1 \beta_1^{2t} + \delta_2 \beta_2^{2t} + \delta_3 \beta_3^{2t} + \dots + \delta_e \beta_e^{2t} \end{aligned} \quad (16)$$

There is a lot of decoding redundancy and delay caused by the RSC decoding method with constant parameters in a time-varying channel because the signal-to-noise ratio (SNR) fluctuates quickly with time. Fig. 3 shows the flow diagram for our proposed new RSC decoding method in method 1, which aims to decrease the total amount of TVs and symptoms. In order to ensure effective decoding, the algorithm dynamically adjusts the total number of TVs and syndromes by evaluating the error symbol amount of each codeword through counting the number of unreliable bits [28].

Table 1

The correlation between the decoding parameters and the counter value of the received RS (255,239) code.

Counter	2	4	6	8	10	12	14	16	24	36	>36
TV num. ζ	1	1	1	1	1	1	1	1	4	8	16
ta	1	2	3	4	5	6	7	8	8	8	8
Error num.	1	2	3	4	5	6	7	8	9	10	>10

The error vector is obtained by calculating just the first two syndromes when $e < t$. Reducing the total number of equations to be evaluated from $2t$ to $2e$ may help in reducing the need to calculate $2t-2e$ equations twice. We should maximize the use of the channel's soft information to get a broader decoding radius if counter is greater than $Th2$. An outcome of statistical regulation is a counter that may be used to calculate the chosen number of televisions. Furthermore, we provide two TV selection while sorting algorithms to identify the TV that can be decoded effectively with less trials. To get the TV index set $\alpha = \{TV1, TV2, \dots, TV\}$, we need to locate $TV2$ using the decoding complementarity principle, then find $TV3$ using the decoding of the two previous TVs and so on. We need to keep doing these steps until we have all of the TVs. According to, the likelihood of the symbol mistake having a 1-bit error is >99 % when the SNR surpasses 6.1 dB.

The first syndrome is $S_0 = R(\alpha^0) = \sum_{l=1}^e E_l$, E_l is the error vector and $R(x)$ is the received codeword. Per the 1 included in the parity number S_0 two sets of indexes, an odd index set, and the concept of compensating decoding $\Gamma_o = \{TV_{o1}, TV_{o2}, \dots, TV_{o\zeta}\}$ and even index set $\Gamma_e = \{TV_{e1}, TV_{e2}, \dots, TV_{e\zeta}\}$, are examined. A total of sixteen TVs were used to decode 255,239 RS codes in the simulation, with each TV serving as the optimal replacement for the ones that came before it. Furthermore, we provide the appropriate optimum test order for both odd and even numbers of 1 in S_0 .

4.1. ResNet model

As a member of the ResNet family, ResNet-152 is a deep residual network algorithm that helped popularize the idea of residual learning, a game-changer for training ultra-deep neural networks. Despite having 152 levels, it successfully overcomes the vanishing gradient issue by skipping over some layers via skip connections or shortcuts. The architecture's remarkable capability at scaling to hundreds of layers makes it a popular choice for image classification jobs.

With an emphasis on maximizing the latency-accuracy trade-off, MobileNetV2 is built for situations that are both mobile and limited in resources. Lightweight depthwise convolutions are used to efficiently filter features, and an inverted residual structure is introduced when the residual connections lie between the bottleneck layers. Without significantly sacrificing accuracy, this model has a compact size and minimal computing needs.

Another member of the ResNet family, ResNet-50 is shallower than ResNet-152 but still quite deep (50 layers). It uses the same ideas of residual learning to make deep network training easier by letting users reuse features and stopping training from becoming worse. ResNet-50 is a strong option for academic with industrial applications because of its great ratio of depth, performance, and computing efficiency. It is used across many areas.

One member of the EfficientNet family, EfficientNetV2 is well-known for its ability to scale models in a way that maintains a good balance between depth, breadth, and resolution. This specific version, "V2," is fine-tuned for practical use, with an eye on improving accuracy and efficiency across a range of hardware platforms. This architecture is ideal for applications that need great computing performance and accuracy because it uses compound scaling with a modified version of scaling models to improve training speed with parameter efficiency.

To make better use of scarce computational resources, the attention mechanism mimics the way the human visual nerve processes data. Different types of attention may be categorized based on their use: spatial (feature) attention, temporal attention, etc. Combinations using encoder-decoder frameworks, like Transformer, are common.

As seen in Fig. 2, the general attention process may be depicted as a key-value pair. To begin, find the distribution of attention $A = \{a_{ij}\}$ among tensors K and:

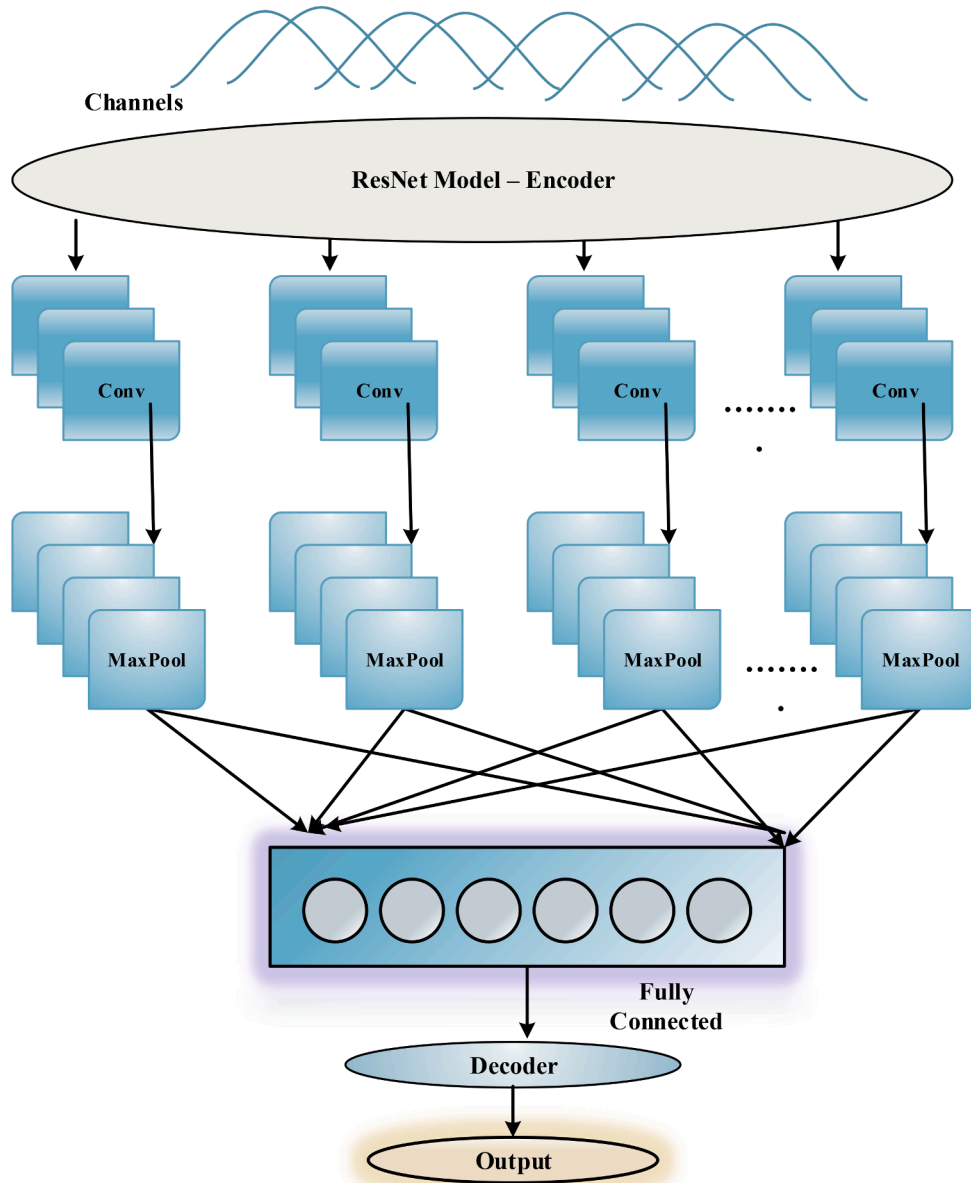


Fig. 3. ResNet based reed solomon code model.

$$\alpha_{if} = \text{softmax}(s(k_f, q_i)) = \frac{\exp(s(k_f, q_i))}{\sum_{m=1}^N \exp(s(k_m, q_i))} \quad (17)$$

where N is the length of the input sequence, $i, j \in \{1, 2, \dots, N\}$ represents the Attention Scoring Function (ASF), which includes scaled dot product, cosine analogy, etc., k_j, q_i denotes the j th row vector of K , and $s(\cdot)$ denotes the locations of the input and output sequences, respectively. In this article, the scaled dot product is used.

You may output H in two different ways. The weighted sum of tensor V , which is dependent on tensor A , may be calculated in one approach as:

$$h_i = \sum_{j=1}^N \alpha_{ij} v_j \quad (18)$$

where h_i represents the j -th row vector of V , and is the row vector of H .

Tensor V and tensor A are multiplied in the alternative approach, which is widely used in self-attention, as:

$$H = A \times V \quad (19)$$

5. Results & discussion

5.1. Implementation results

In this experimental setup, we generate 1000,000 samples, with 10,000 samples kept in int32. The sample's SNR falls uniformly within the range of -10 dB to 10 dB. The channel coding type recognition system was trained using the data set that had 70 % of the samples, and its performance was evaluated using the data set that contained 30 % of the samples. We use an Nvidia 1080 Ti GPU to do all of our testing and training in Keras. During training, we first set the learning rate at 0.002. After each 10 epochs, we increase the study rate to 10 % of the primary if the training loss has not decreased. A training network with 200 epochs trained with a batch size of 16.

This (255,239) RS code time-varying channel adaptive RSC decoder was put into action utilizing a $0.13 \mu\text{m}$ and 65 nm CMOS technology. According to Table 2, using Synopsys design tools, the highest clock frequency for the suggested decoder is 385 MHz in a $0.13 \mu\text{m}$ process as well as 550 MHz in a 65 nm process. The suggested RSC decoder

Table 2
Implementation results of the proposed RSC decoders.

Method	[17]	[19]	[21]	RS Codec	Proposed
Tech.	0.14 μm	68 nm	72 nm	0.17 μm	69 nm
f_{max} (MHz)	240	570	590	399	560
Power (mW/MHz)	-	29.0@590	28.5@570	266.2@395	35.7@550
Throughput (Gbps)	1.8a	4.8a	8.9a	17.7b	27.7b
Gate count (kXORs) with buffer	29.8	35.1	35.9	59.6	58.8
Coding gain (dBs@FER)	0.39@10 ⁻⁷	0.66@10 ⁻⁸	0.70@10 ⁻⁶	0.67@10 ⁻⁶	0.59@10 ⁻⁶
Latency (clock cycles)	505a	537a	288a	57b	60b

a Fixed value b Average value.

provides coding gain that is equal to or greater than that of the state-of-the-art RSC decoders currently available. Power consumption of the suggested decoder is increased due to the elevated clock frequency with extra hardware resources.

On average, 1.06 TVs are needed for the suggested decoder to finish decoding for the time-varying channel having a range of [6.5,8] dB. In comparison to the RSC decoder in reference 11, the decoder's average delay is 81.6 % shorter, at 50 cycles, this is calculated as $17+1.06 \times 16+16$. Formula (5) may be used to determine the average speed of the proposed decoder.

$$\text{Throughput} = \frac{\text{\# of bits processed each time}}{\text{\# of clock cycles needed} \times T_{\min}} \quad (20)$$

The assessed data is $(255 \times 8 \times 385)/50 = 15.7 \text{ Gb/s}$ in $0.13 \mu\text{m}$ process. The estimated throughput is $(255 \times 8 \times 550)/50 = 22.4 \text{ Gb/s}$ in the 65 nm method.

The suggested decoder has a throughput that is 8.8, 4.1, and 1.5 times higher than before. Lower latency and faster throughput compared to the state-of-the-art RSC decoder suggest that the proposed decoder has improved hardware performance.

5.2. Numerical results and performance discussion

The use of CNNs or other deep learning techniques is largely intended to enhance the efficiency of soft-decision decoding for RSC (refer

Table 3
Comparison of metrics with traditional decoders and numerical gain.

Metrics	Comparison to Traditional Decoders	Numerical Gain
Coding Gain (BER/FER)	Greater capacity for error correction, particularly at medium to high SNRs.	Based on the code rate and algorithm, there may be a gain of around 0.3 dB to 1.4 dB over Hard-Decision Decoding (HDD). For example, RS(255, 239) can achieve a gain of up to ~0.75 dB, while RS(63, 55) can achieve a gain of up to ~1.4 dB.
Maximum Likelihood (ML) Gap	Theoretical Maximum Likelihood (ML) limit is closer to the actual decoder performance.	Compared to conventional low-complexity algebraic soft-decision decoders (such as Low-Complexity Chase), CNN-aided decoders are able to attain performance that is more comparable to ML decoding.
Complexity/Throughput	There is always a catch: optimizing neural decoders may provide a high throughput, but improving performance could make training and execution more computationally difficult.	While giving better error rates than some low-complexity algebraic approaches, improved CNN/RNN-based soft decoders have been shown to retain high throughput, reaching up to about 1.19 GBit/s for RS(255, 239).

Table 3). This way of decoding has historically depended on complicated algebraic computations.

Because of its simplicity and speed, hard-decision (HDD) RS decoding is widely used in these systems. Nevertheless, HDD is unable to use the vital "soft" reliability data provided by the channel demodulator, resulting in a decrease in performance, typically around -2 to -3 dB on an AWGN channel. To make up for this lost coding benefit, researchers are looking at deep learning decoders like CNNs to build a simple yet fast soft-decision decoder (SDD). In conclusion, numerical results demonstrate that RS decoders can greatly benefit from CNN-like structures when it comes to error correction. This integration leads to a measurable coding gain in dB for the same bit error rate (BER), which could pave the way for more reliable reception even in difficult broadcast environments, such as areas with weak signals or mobile reception [29,30].

The CNN-assisted decoder makes good use of the demodulator's soft-information, which consists of channel dependability data. When compared to the conventional decoders that are still in use by many DVB standards (such as DVB-T/H), numerical findings show a substantial improvement in coding gain (usually ranging from 0.3 dB to >1.0 dB) in Bit Error Rate (BER) or Frame Error Rate (FER). This improvement immediately leads to better reception even when the channel is not ideal.

- **Optimal Soft-Decision Decoding:** The CNN-based method can learn the intricate non-linear relationships in the error patterns, allowing it to achieve near-optimal soft-decision decoding performance with minimal computational overhead, all while keeping a structure that is conducive to efficient hardware implementation.
- **DVB/Digital Media Real-World Use:** In many broadcasting situations (e.g., mobile DVB-H, portable digital radio), the improved decoding capabilities makes the system more resistant to spurious signals, burst mistakes, and low Signal-to-Noise Ratio (SNR). By making links more reliable and decreasing the occurrence of "cliff effects" (i.e., abrupt loss of signal), this enhances QoS directly.
- **While CNNs provide better performance,** researchers still face the major difficulty of developing decoder architectures with low complexity while simultaneously controlling the training phase overhead. To meet the high-throughput demands of digital broadcasting without incurring excessive hardware costs, future research will concentrate on lightweight network topologies (such as pruning, quantization, and weight sharing). Deep learning decoders have a long way to go before they can compete with traditional hardware in terms of efficiency and performance.

6. Conclusion

In this research, we developed a time-varying channel adaptive RSC method that can identify channels and dynamically modify decoding parameters accordingly. Decoding complexity with latency may be reduced using the suggested technique by drastically cutting down on important equations and redundant TVs. We used effect evaluation in the time-varying channel to test how well the technique worked. There was an 81.6 % decrease in average decoder delay and a 50.4 % decrease in the number of TVs needed when compared to the state-of-the-art RSC

algorithm. A time-varying channel RSC decoder based on the aforementioned methods is detailed below, along with its hardware design. The implementation findings demonstrate that the suggested RSC decoder produced a throughput gain of at least 150 % when compared with previously published studies. We will be concentrating on time-varying RSC decoders with great performance in the future. For future research, we need to address the non-binary nature of RS codes and to minimize computational complexity (FLOPS) and memory footprint for FPGA or ASIC implementation, we may develop CNN models that are severely pruned, quantized, or binarized. This will help create low-complexity CNN architectures. Constructing unfolded or recurrent architectures with shared weights across symbols or iterations to reduce the overall amount of parameters that may be trained. High-Throughput Implementation: o Develop CNN-specific parallel computing architectures (e.g., using systolic arrays or dedicated tensor cores) to meet the gigabit-per-second throughput demands

Ethics approval and consent to participate

I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country.

Consent for publication

I confirm that any participants (or their guardians if unable to give informed consent, or next of kin, if deceased) who may be identifiable through the manuscript (such as a case report), have been given an opportunity to review the final manuscript and have provided written consent to publish.

Availability of data and materials

The data used to support the findings of this study are available from the corresponding author upon request.

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CRedit authorship contribution statement

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Declaration of competing interest

Here are no have no conflicts of interest to declare. All authors have seen and agree with the contents of the manuscript and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication.

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Data availability

No data was used for the research described in the article.

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